

WEBCAM MICROSCOPE (₹2,325)



Your average webcam can easily be repurposed to act as a microscope. By simply moving the objective lens away from the sensor you can gain magnification levels. What remains to be done is to provide a mechanism that can accurately move your along a straight axis and keep it stable. Here's how we did it.

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MATERIALS NEEDED

- 1x Dowel Rods: ₹15
- Spoilt DVD-ROM drive: Free
- Cardboard - Free
- Webcam: ₹1800
- Glue gun: ₹500
- Opaque cellophane tape: ₹10



STEP 01 We're using an old CD-ROM drive's tray assembly to act as the slide mechanism. Take the drive apart till you can easily move the tray in or out. Ensure that it stays in a position once you let go.



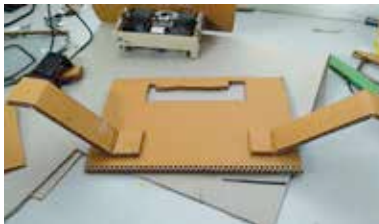
STEP 02 Start off by tracing the base of the CD-ROM drive (the side with the power and data connectors) onto a corrugated cardboard square or rectangle. Then cut out that particular piece. Use a plain thick cardboard and layer it underneath the sheet to add weight. This is the base.



STEP 03 Take another similar sized cardboard piece and cut out the same slot as you did previously. This board is the upper plate.



STEP 04 On the second piece that you just cut out, add another slit that's about 6 cms wide and 3 cms deep. This slit should be 1 cm away from the previous slit.



STEP 05 Take two rectangular pieces of corrugated cardboard and glue them onto the base to build pillars in order to hold the two sheets one above the other like a table.



STEP 06 Slide the CD-ROM drive into the upper plate and ensure it is a snug fit. Don't superglue the upper plate to the CD-ROM drive yet.



STEP 07 Use superglue or hot-glue to place the upper plate with the CD-ROM drive in it, onto the base plate while carefully aligning the slits. Your CD-ROM drive should remain stationary without sliding around.



STEP 08 Take your webcam and check out YouTube videos to figure out how to disassemble it.



STEP 09 Once you have the webcam's faceplate removed, you need to unscrew the objective lens. It's the shiny part on the left of the webcam as seen here.